13 Fourier's law, series and transform

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Learning outcomes

You should be able to:

- Discuss the link between Fourier's law of heat conduction, the Fourier series and the Fourier transform.
- Transform a partial differential equation (PDE) to a set of ordinary differential equations (ODE) using separation of variables.
- Constrain the boundary condition of a PDE to an arbitrary spatial distribution of a state variable using an integral transform.
- Simplify trigonometric functions evaluated at integer values.
- Integrate products of piecewise periodic functions with trigonometric functions and other polynomials.

13.1 Fourier's law of heat conduction

Joseph Fourier was a 18th/19th century French mathematician. There are many mathematical subjects associated with Fourier. Three of note are Fourier's law, the Fourier series and the Fourier transform.

Fourier's law is the idea that heat conduction can be modeled using a diffusion equation of the form

$$\rho c_p \frac{\partial T}{\partial t} - \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) - \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) - \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) = 0$$

where ρ [$\mathrm{ML^{-3}}$] is density, c_p [$\mathrm{L^2T^{-2}\Theta^{-1}}$] is constant-pressure specific heat capacity, T [Θ] is temperature, t [T] is time and k [$\mathrm{MLT^{-3}\Theta^{-1}}$] is thermal conductivity.

Now consider two dimensional steady state heat conduction in a homogenous medium:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \tag{1}$$

where the following boundary conditions apply:

$$\begin{array}{lll} T = 0, & x = 0, & 0 \leq y \leq b \\ T = 0, & x = a, & 0 \leq y \leq b \\ T = 0, & 0 \leq x \leq a, & y = 0 \\ T = f(x), & 0 \leq x \leq a, & y = b \end{array}$$

and a [L] and b [L] are the breadth and length of the domain, respectively and $f(x)$ [Θ] is an arbitrary function of x.

In what follows, we will transform the PDE in Eq. (1) into two ODEs using separation of variables. The three zero temperature boundaries are easily enforced by determining values for the integration constants. The $f(x)$ boundary is then imposed using an innovative integration method developed by Fourier, which

gave rise to the development of the Fourier series and Fourier transform.

Although the original motivation of the Fourier series was to solve PDEs, the series has become invaluable in the field of signal processing and inverse modeling.

Challenge 1a

Consider the partial differential equation for two-dimensional steady-state heat conduction

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Substitute $T(x,y) = F(x)G(y)$ and separate the variables to derive two associated ordinary differential equations.

Solution

Making the specified substitution leads to

$$\frac{\partial^2 (FG)}{\partial x^2} + \frac{\partial^2 (FG)}{\partial y^2} = 0$$

Because G is constant in x it can come outside of the partial derivative with respect to x. Similarly, because F is constant in y it can come outside of the partial derivative with respect to y. It follows that

$$G\frac{\partial^2 F}{\partial x^2} + F\frac{\partial^2 G}{\partial y^2} = 0$$

To separate the variables, we need to get all the x factors on one side and the y factors on the other side:

$$\frac{1}{G}\frac{\partial^2 G}{\partial y^2} = -\frac{1}{F}\frac{\partial^2 F}{\partial x^2}$$

Note that the left-hand-side of the above equation is purely a function of x whereas the right-hand-side is purely a function of y. It follows that both sides are therefore actually constant in both x and y such that it can be said that

$$\frac{1}{G}\frac{\partial^2 G}{\partial y^2} = -\frac{1}{F}\frac{\partial^2 F}{\partial x^2} = \alpha$$

where α is a constant yet to be defined.

In this way, two ODEs are derived:

$$\frac{\partial^2 F}{\partial x^2} + \alpha F = 0 \tag{2}$$

$$\frac{\partial^2 G}{\partial y^2} - \alpha G = 0 \tag{3}$$

Challenge 1b

Derive general solutions for F and G.

Solution

Eq. (2) represents a linear and homogenous second-order differential equation. The associated auxiliary equation takes the form

$$\lambda^2 + \alpha = 0$$

which has imaginary roots $\lambda = \pm i\sqrt{\alpha}$. It follows that the general solution for Eq. (2) is

$$F = A\cos(\sqrt{\alpha}x) + B\sin(\sqrt{\alpha}x)$$

where A and B are integration constants yet to be defined.

Eq. (3) also represents a linear and homogenous second-order differential equation. The associated auxiliary equation takes the form

$$\lambda^2 - \alpha = 0$$

which has the roots $\lambda = \pm\sqrt{\alpha}$. It follows that the general solution for Eq. (2) is

$$G = C\cosh(\sqrt{\alpha}y) + D\sinh(\sqrt{\alpha}y)$$

where C and D are another set of integration constants yet to be defined.

Challenge 1c

Given the general solutions for F and G apply the following boundary conditions:

$$\begin{array}{lll} T = 0, & x = 0, & 0 \leq y \leq b \\ T = 0, & x = a, & 0 \leq y \leq b \\ T = 0, & 0 \leq x \leq a, & y = 0 \end{array}$$

Solution

Recall that $T(x,y) = F(x)G(y)$ where

$$F = A\cos(\sqrt{\alpha}x) + B\sin(\sqrt{\alpha}x)$$

$$G = C\cosh(\sqrt{\alpha}y) + D\sinh(\sqrt{\alpha}y)$$

First consider the boundary at $x = 0$. We need to choose a value of A or B such that $F = 0$ when $x = 0$. Noting that $\sin 0 = 0$ it follows that we should choose A to be zero. Therefore

$$F = B\sin(\sqrt{\alpha}x)$$

Now we will consider the boundary at $y = 0$. We need to choose a value for C or D such that $G = 0$ when $y = 0$. Noting that $\sinh 0 = 0$ it follows that we should choose C to be zero. Therefore

$$G = D\sinh(\sqrt{\alpha}y)$$

Given that $T(x,y) = F(x)G(y)$ we can now say that

$$T(x,y) = E\sin(\sqrt{\alpha}x)\sinh(\sqrt{\alpha}y)$$

where $E = BD$.

Now we will consider the boundary at $x = a$. We need to choose a value for E or α such that $T = 0$ when $x = a$. It is clear that there is no non-zero value of E that can satisfy this condition. Therefore we need to choose a value of α such that $\sin(\sqrt{\alpha}a) = 0$. Noting that $\sin(n\pi) = 0$ where n is an integer, we should choose

$$\sqrt{\alpha} = \frac{n\pi}{a}$$

such that we have

$$T(x,y) = E\sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi y}{a}\right) \tag{4}$$

13.2 Non-uniform boundary conditions

Unfortunately the boundary condition $T(x, y = b) = f(x)$ is less straightforward to apply. The problem is that it is not possible to choose a constant value of E such that $T = f(x)$ for $0 \leq x \leq a$. However, Fourier recognized that $n = 1, 2, 3, \ldots, \infty$. Therefore, Eq. (4) represents a set of solutions:

$$T_n(x,y) = E_n\sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi y}{a}\right)$$

where E_n represents a set of integration constants yet to be defined.

It is therefore possible to write a solution as a summation of these alternative solutions, i.e.,

$$T(x,y) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right)$$

Challenge 2a

Consider the solution

$$T(x,y) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right) \tag{5}$$

Apply the boundary condition $T(x,b) = f(x)$ and derive an expression for $f(x)$ in terms of E_n.

Solution

The boundary condition implies that $T = f(x)$ when $y = b$. It follows that

$$f(x) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi b}{a}\right) \tag{6}$$

Challenge 2b

Multiply both sides of your result from Challenge 2a by $\sin(m\pi x/a)$ (where m is another set of positive integers) and integrate both sides with respect to x from 0 to a.

Solution

$$\begin{aligned} &\int_0^a f(x) \sin\left(\frac{m\pi x}{a}\right) dx \\ &= \sum_{n=1}^{\infty} E_n \sinh\left(\frac{n\pi b}{a}\right) \int_0^a \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx \end{aligned} \tag{7}$$

To deal with the integral on the right-hand-side, first invoke the relationship

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

from which it can be said that

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

and consequently

$$\sin A \sin B = \frac{\cos(A-B) - \cos(A+B)}{2}$$

It follows that

$$\begin{aligned}
&\int_0^a \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi x}{a}\right) dx \\
&= \int_0^a \frac{1}{2}\left[\cos\left(\frac{(n-m)\pi x}{a}\right) - \cos\left(\frac{(n+m)\pi x}{a}\right)\right] dx \\
&= \frac{a}{2\pi}\left[\frac{\sin\left(\frac{(n-m)\pi x}{a}\right)}{n-m} - \frac{\sin\left(\frac{(n+m)\pi x}{a}\right)}{n+m}\right]_0^a \qquad (8) \\
&= \frac{a}{2\pi}\left[\frac{\sin\left((n-m)\pi\right)}{n-m} - \frac{\sin\left((n+m)\pi\right)}{n+m}\right]
\end{aligned}$$

Substituting Eq. (8) into Eq. (7) then yields

$$\begin{aligned}
&\int_0^a f(x) \sin\left(\frac{m\pi x}{a}\right) dx \\
&= \sum_{n=1}^{\infty} E_n \sinh\left(\frac{n\pi b}{a}\right) \frac{a}{2\pi}\left[\frac{\sin\left((n-m)\pi\right)}{n-m} - \frac{\sin\left((n+m)\pi\right)}{n+m}\right]
\end{aligned}$$

But note that because n and m are all positive integers, all the terms on the right-hand-side are zero except for the ones where $n = m$. Recall that

$$\lim_{\varepsilon \to 0} \frac{\sin(\varepsilon x)}{\varepsilon} = x$$

It follows that

$$\int_0^a f(x) \sin\left(\frac{m\pi x}{a}\right) dx = E_m \sinh\left(\frac{m\pi b}{a}\right) \frac{a}{2} \tag{9}$$

Challenge 2c

Use your result from Challenge 2b to determine E_n and write a solution for $T(x,y)$ that satisfies the boundary condition, $T(x,b) = f(x)$.

Solution

Solving Eq. (9) for E_m leads to

$$E_m = \frac{2}{a \sinh(m\pi b/a)} \int_0^a f(x) \sin(m\pi x/a) dx$$

If we set $m = n$ we will therefore have

$$E_n = \frac{2}{a \sinh(n\pi b/a)} \int_0^a f(x) \sin(n\pi x/a) dx \tag{10}$$

which on substitution back into Eq. (5) leads to

$$T(x,y) = \sum_{n=1}^{\infty} \frac{2 \sin(n\pi x/a) \sinh(n\pi y/a)}{a \sinh(n\pi b/a)} \int_0^a f(x) \sin(n\pi x/a) dx \tag{11}$$

Challenge 2d

Use your result from Challenge 2c to write a solution for $T(x,y)$ when

$$f(x) = \begin{cases} T_0\left(\dfrac{x}{c}\right), & 0 \le x \le c \\ \\ T_0\left(\dfrac{a-x}{a-c}\right), & c < x \le a \end{cases} \tag{12}$$

Solution

Let

$$F_n = \int_0^a f(x)\sin(n\pi x/a)dx \tag{13}$$

such that

$$T(x,y) = \sum_{n=1}^{\infty} \frac{2F_n \sin(n\pi x/a)\sinh(n\pi y/a)}{a\sinh(n\pi b/a)}$$

Substituting Eq. (12) into Eq. (13) leads to

$$F_n = \int_0^c T_0\left(\frac{x}{c}\right)\sin(n\pi x/a)dx + \int_c^a T_0\left(\frac{a-x}{a-c}\right)\sin(n\pi x/a)dx \tag{14}$$

It can be shown that:

$$\int_{x_1}^{x_2} \sin(n\pi x/a)dx = \left[-\frac{\cos(n\pi x/a)}{n\pi/a}\right]_{x_1}^{x_2}$$

and

$$\int_{x_1}^{x_2} x\sin(n\pi x/a)dx = \left[\frac{\sin(n\pi x/a)}{(n\pi/a)^2} - \frac{x\cos(n\pi x/a)}{n\pi/a}\right]_{x_1}^{x_2}$$

and therefore

$$\int_{x_1}^{x_2} (a-x)\sin(n\pi x/a)dx = \left[\frac{(x-a)\cos(n\pi x/a)}{n\pi/a} - \frac{\sin(n\pi x/a)}{(n\pi/a)^2}\right]_{x_1}^{x_2}$$

It follows that

$$\int_0^c \left(\frac{x}{c}\right)\sin(n\pi x/a)dx = \frac{1}{c}\left[\frac{\sin(n\pi x/a)}{(n\pi/a)^2} - \frac{x\cos(n\pi x/a)}{n\pi/a}\right]_0^c$$

$$= \frac{\sin(n\pi c/a)}{c(n\pi/a)^2} - \frac{\cos(n\pi c/a)}{n\pi/a}$$

and

$$\int_c^a \left(\frac{a-x}{a-c}\right)\sin(n\pi x/a)dx = \left[\frac{(x-a)\cos(n\pi x/a)}{(a-c)n\pi/a} - \frac{\sin(n\pi x/a)}{(a-c)(n\pi/a)^2}\right]_c^a$$

$$= \frac{\sin(n\pi c/a)}{(a-c)(n\pi/a)^2} + \frac{\cos(n\pi c/a)}{n\pi/a}$$

Substituting the above two expressions into Eq. (14) then leads to

$$F_n = T_0\left[\frac{\sin(n\pi c/a)}{c(n\pi/a)^2} - \frac{\cos(n\pi c/a)}{n\pi/a} + \frac{\sin(n\pi c/a)}{(a-c)(n\pi/a)^2} + \frac{\cos(n\pi c/a)}{n\pi/a}\right]$$

$$= \frac{T_0 a^3 \sin(n\pi c/a)}{c(a-c)(n\pi)^2}$$

13.3 The Fourier sine series and sine transform

But consider again Eqs. (6) and (10):

$$f(x) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{n\pi x}{a}\right)\sinh\left(\frac{n\pi b}{a}\right)$$

$$E_n = \frac{2}{a\sinh(n\pi b/a)}\int_0^a f(x)\sin(n\pi x/a)dx$$

Letting $b_n = E_n \sinh(n\pi b/a)$ suggests that a continuous function, $f(x)$, can be approximated by a summation of sine waves multiplied by a set of weighting coefficients, b_n, i.e.,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{a}\right) \tag{15}$$

where the weighting coefficients are determined from

$$b_n = \frac{2}{a}\int_0^a f(x)\sin(n\pi x/a)dx \tag{16}$$

The expansion in Eq. (15) is known as the Fourier sine series and Eq. (16) is known as the Fourier sine transform.

Further questions

1) Simplify the following expressions as much as possible assuming that n is an integer. Take care to consider if more simplifications are possible if n is odd or even.

a) $\sin(n\pi)$ b) $\cos(n\pi)$ c) $\cos^2(n\pi)$

d) $1+\cos(n\pi)$ e) $1-\cos(n\pi)$

2) Integrate the following functions with respect to x.

a) $\sin(nx)$ b) $\cos(nx)$ c) $x\sin(nx)$

d) $x\cos(nx)$ e) $x^2\sin(nx)$ f) $x^2\cos(nx)$

3) Sketch the following piecewise period functions for x between -2π and 4π assuming that each function has a period of 2π.

a) $f(x) = x-\pi, \quad 0 \le x \le 2\pi$

b) $f(x) = x(2\pi - x), \quad 0 \le x \le 2\pi$

c) $f(x) = \begin{cases} x, & 0 \le x < \pi \\ 2\pi - x, & \pi \le x \le 2\pi \end{cases}$

d) $f(x) = \begin{cases} -1, & 0 \le x < \pi \\ 1, & \pi \le x \le 2\pi \end{cases}$

4) Consider the following notation

$$\langle \cdot \rangle = \frac{1}{\pi} \int_0^{2\pi} \cdot dx$$

such that, for example,

$$\langle f(x) \cos x \rangle = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos x dx$$

For each of the functions in Question 3, determine the following definite integrals, where n (if present) is always an integer value greater than zero.

i) $\langle f(x) \rangle$ ii) $\langle f(x)x \rangle$

iii) $\langle f(x) \sin(nx) \rangle$ iv) $\langle f(x) \cos(nx) \rangle$

5) Consider again the $\langle \cdot \rangle$ notation defined in Question 4 above. Assuming that m and n are integers, determine the following definite integrals with special consideration of the cases $m = n$, $m \neq n$, $m = 0$ and $n = 0$.

a) $\langle \sin(nx) \sin(mx) \rangle$ b) $\langle \sin(nx) \cos(mx) \rangle$

c) $\langle \cos(nx) \cos(mx) \rangle$

Solutions to further questions

1a) $\sin(n\pi) = 0$

b) For even values of n, $\cos(n\pi) = 1$, but for odd values of n, $\cos(n\pi) = -1$.

This can be compactly stated by saying $\cos(n\pi) = (-1)^n$

c) From the above discussion it is then clear that $\cos^2(n\pi) = 1$

d) From part b) above $1 + \cos(n\pi) = 1 + (-1)^n$

e) From part b) above $1 - \cos(n\pi) = 1 - (-1)^n = 1 + (-1)^{n+1}$

2)

a) $$\int \sin(nx)dx = -\frac{\cos(nx)}{n} + C$$

b) $$\int \cos(nx)dx = \frac{\sin(nx)}{n} + C$$

c) $$\int x\sin(nx)dx = -\frac{x\cos(nx)}{n} + \int \frac{\cos(nx)}{n}dx$$

$$= -\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2} + C$$

d) $$\int x\cos(nx)dx = \frac{x\sin(nx)}{n} - \int \frac{\sin(nx)}{n}dx$$

$$= \frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2} + C$$

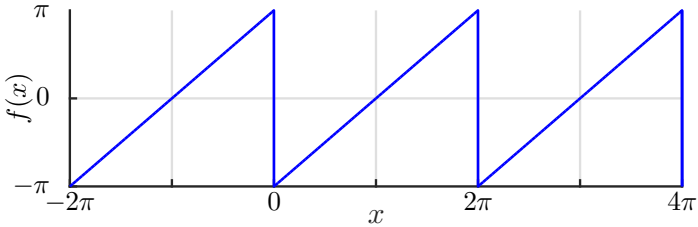
e) $$\int x^2 \sin(nx)dx = -\frac{x^2\cos(nx)}{n} + \frac{2}{n}\int x\cos(nx)dx$$

$$= -\frac{x^2\cos(nx)}{n} + \frac{2x\sin(nx)}{n^2} + \frac{2\cos(nx)}{n^3} + C$$

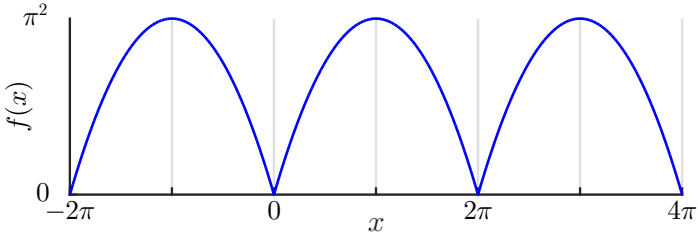
f) $$\int x^2 \cos(nx)dx = \frac{x^2\sin(nx)}{n} - \frac{2}{n}\int x\sin(nx)dx$$

$$= \frac{x^2\sin(nx)}{n} + \frac{2x\cos(nx)}{n^2} - \frac{2\sin(nx)}{n^3} + C$$

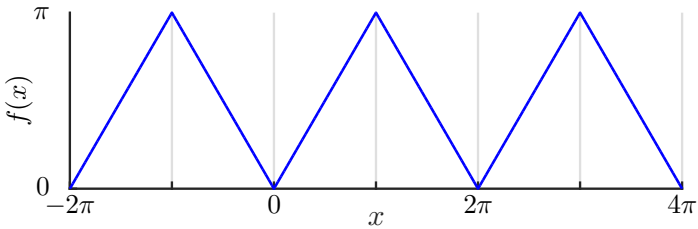
3a) Sketch of $f(x) = x - \pi, \quad 0 \le x \le 2\pi$



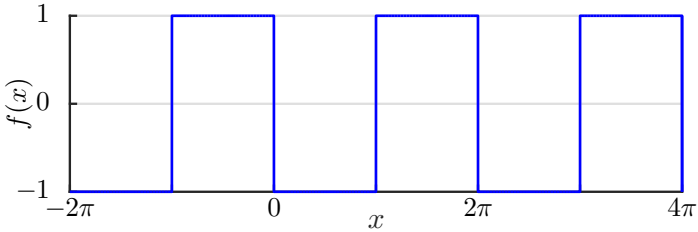
b) Sketch of $f(x) = x(2\pi - x), \quad 0 \le x \le 2\pi$



c) Sketch of $f(x) = \begin{cases} x, & 0 \le x < \pi \\ 2\pi - x, & \pi \le x \le 2\pi \end{cases}$



d) Sketch of $f(x) = \begin{cases} -1, & 0 \le x < \pi \\ 1, & \pi \le x \le 2\pi \end{cases}$



4a) $f(x) = x - \pi$

(i) $\dfrac{1}{\pi}\displaystyle\int_0^{2\pi}(x-\pi)dx$

$$= \left[\frac{x^2}{2\pi} - x\right]_0^{2\pi} = 2\pi - 2\pi = 0$$

(ii) $\dfrac{1}{\pi}\displaystyle\int_0^{2\pi}(x-\pi)xdx$

$$= \left[\frac{x^3}{3\pi} - \frac{x^2}{2}\right]_0^{2\pi} = \frac{8\pi^2}{3} - \frac{4\pi^2}{2} = \frac{2\pi^2}{3}$$

(iii) $\dfrac{1}{\pi}\displaystyle\int_0^{2\pi}(x-\pi)\sin(nx)dx$

$$= \frac{1}{\pi}\left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2} + \frac{\pi\cos(nx)}{n}\right]_0^{2\pi}$$

$$= -\frac{\cos(2n\pi)}{n} + \frac{\cos(2n\pi)}{n} - \frac{1}{n} = -\frac{2}{n}$$

(iv) $\dfrac{1}{\pi}\displaystyle\int_0^{2\pi}(x-\pi)\cos(nx)dx$

$$= \frac{1}{\pi}\left[\frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2} - \frac{\pi\sin(nx)}{n}\right]_0^{2\pi}$$

$$= \frac{\cos(2n\pi)}{n^2\pi} - \frac{1}{n^2\pi} = 0$$

4b) $f(x) = x(2\pi - x)$

(i) $$\frac{1}{\pi}\int_0^{2\pi} x(2\pi - x)dx = \left[x^2 - \frac{x^3}{3\pi}\right]_0^{2\pi} = 4\pi^2 - \frac{8}{3}\pi^2 = \frac{4\pi^2}{3}$$

(ii) $$\frac{1}{\pi}\int_0^{2\pi} x(2\pi - x)xdx = \left[\frac{2x^3}{3\pi} - \frac{x^4}{4\pi}\right]_0^{2\pi} = \frac{16\pi^3}{3} - \frac{16\pi^3}{4} = \frac{4\pi^3}{3}$$

(iii) $$\frac{1}{\pi}\int_0^{2\pi} x(2\pi - x)\sin(nx)dx = \frac{1}{\pi}\left[2\pi\left(\frac{\sin(nx)}{n^2} - \frac{x\cos(nx)}{n}\right)\right.$$
$$\left. + \left(\frac{x^2\cos(nx)}{n} - \frac{2x\sin(nx)}{n^2} - \frac{2\cos(nx)}{n^3}\right)\right]_0^{2\pi}$$

$$= \left[\left(2 - \frac{2x}{\pi}\right)\frac{\sin(nx)}{n^2} - \left(2x - \frac{x^2}{\pi} + \frac{2}{n^2\pi}\right)\frac{\cos(nx)}{n}\right]_0^{2\pi}$$

$$= -\left(4\pi - \frac{4\pi^2}{\pi} + \frac{2}{n^2}\pi\right)\frac{1}{n} + \left(\frac{2}{n^2}\pi\right)\frac{1}{n} = 0$$

(iv) $$\frac{1}{\pi}\int_0^{2\pi} x(2\pi - x)\cos(nx)dx = \frac{1}{\pi}\left[2\pi\left(\frac{x\sin(nx)}{n} + \frac{\cos(nx)}{n^2}\right)\right.$$
$$\left. - \left(\frac{x^2\sin(nx)}{n} + \frac{2x\cos(nx)}{n^2} - \frac{2\sin(nx)}{n^3}\right)\right]_0^{2\pi}$$

$$= \left[\left(2x - \frac{x^2}{\pi} + \frac{2}{n^2\pi}\right)\frac{\sin(nx)}{n} + \left(2 - \frac{2x}{\pi}\right)\frac{\cos(nx)}{n^2}\right]_0^{2\pi}$$

$$= \left(2 - \frac{4\pi}{\pi}\right)\frac{1}{n^2} - (2)\frac{1}{n^2} = -\frac{4}{n^2}$$

4c) $f(x) = \begin{cases} x, & 0 \le x < \pi \\ 2\pi - x, & \pi \le x \le 2\pi \end{cases}$

(i) $\langle f(x) \rangle = \frac{1}{\pi}\int_0^{\pi} x dx + \frac{1}{\pi}\int_{\pi}^{2\pi} (2\pi - x) dx$

$$= \left[\frac{x^2}{2\pi}\right]_0^{\pi} + \left[2x - \frac{x^2}{2\pi}\right]_{\pi}^{2\pi} = \frac{\pi}{2} + 4\pi - \frac{4\pi}{2} - 2\pi + \frac{\pi}{2} = \pi$$

(ii) $\langle f(x)x \rangle = \frac{1}{\pi}\int_0^{\pi} x^2 dx + \frac{1}{\pi}\int_{\pi}^{2\pi} (2\pi x - x^2) dx$

$$= \left[\frac{x^3}{3\pi}\right]_0^{\pi} + \left[x^2 - \frac{x^3}{3\pi}\right]_{\pi}^{2\pi} = \frac{\pi^2}{3} + 4\pi^2 - \frac{8\pi^2}{3} - \pi^2 + \frac{\pi^2}{3} = \pi^2$$

(iii) $\langle f(x)\sin(nx) \rangle = \frac{1}{\pi}\int_0^{\pi} x\sin(nx) dx + \frac{1}{\pi}\int_{\pi}^{2\pi} (2\pi - x)\sin(nx) dx$

$$= \left[\frac{\sin(nx)}{n^2\pi} - \frac{x\cos(nx)}{n\pi}\right]_0^{\pi} + \left[-\frac{2\cos(nx)}{n} - \frac{\sin(nx)}{n^2\pi} + \frac{x\cos(nx)}{n\pi}\right]_{\pi}^{2\pi}$$

$$= -\frac{(-1)^n}{n} - \frac{2(-1)^{2n}}{n} + \frac{2(-1)^{2n}}{n} + \frac{2(-1)^n}{n} - \frac{(-1)^n}{n} = 0$$

(iv) $\langle f(x)\cos(nx) \rangle = \frac{1}{\pi}\int_0^{\pi} x\cos(nx) dx + \frac{1}{\pi}\int_{\pi}^{2\pi} (2\pi - x)\cos(nx) dx$

$$= \left[\frac{\cos(nx)}{n^2\pi} + \frac{x\sin(nx)}{n\pi}\right]_0^{\pi} + \left[\frac{2\sin(nx)}{n} - \frac{\cos(nx)}{n^2\pi} - \frac{x\sin(nx)}{n\pi}\right]_{\pi}^{2\pi}$$

$$= \frac{(-1)^n}{n^2\pi} - \frac{1}{n^2\pi} - \frac{(-1)^{2n}}{n^2\pi} + \frac{(-1)^n}{n^2\pi} = \frac{2[(-1)^n - 1]}{n^2\pi}$$

4d) $f(x) = \begin{cases} -1, & 0 \le x < \pi \\ 1, & \pi \le x \le 2\pi \end{cases}$

(i) $\langle f(x) \rangle = -\frac{1}{\pi}\int_0^{\pi} 1dx + \frac{1}{\pi}\int_{\pi}^{2\pi} 1dx$

$= -\left[\frac{x}{\pi}\right]_0^{\pi} + \left[\frac{x}{\pi}\right]_{\pi}^{2\pi} = -1 + 2 - 1 = 0$

(ii) $\langle f(x)x \rangle = -\frac{1}{\pi}\int_0^{\pi} xdx + \frac{1}{\pi}\int_{\pi}^{2\pi} xdx$

$= -\left[\frac{x^2}{2\pi}\right]_0^{\pi} + \left[\frac{x^2}{2\pi}\right]_{\pi}^{2\pi} = -\frac{\pi}{2} + \frac{4\pi}{2} - \frac{\pi}{2} = \pi$

(iii) $\langle f(x)\sin(nx) \rangle = -\frac{1}{\pi}\int_0^{\pi} \sin(nx)dx + \frac{1}{\pi}\int_{\pi}^{2\pi} \sin(nx)dx$

$= \left[\frac{\cos(nx)}{n\pi}\right]_0^{\pi} + \left[-\frac{\cos(nx)}{n\pi}\right]_{\pi}^{2\pi}$

$= \frac{(-1)^n - 1}{n\pi} - \frac{[(-1)^{2n} - (-1)^n)}{n\pi} = \frac{2[(-1)^n - 1]}{n\pi}$

(iv) $\langle f(x)\cos(nx) \rangle = -\frac{1}{\pi}\int_0^{\pi} \cos(nx)dx + \frac{1}{\pi}\int_{\pi}^{2\pi} \cos(nx)dx$

$= -\left[\frac{\sin(nx)}{n\pi}\right]_0^{\pi} + \left[\frac{\sin(nx)}{n\pi}\right]_{\pi}^{2\pi} = 0$

5a) $\langle \sin(nx)\sin(mx)\rangle = \dfrac{1}{\pi}\displaystyle\int_0^{2\pi} \sin(nx)\sin(mx)dx$

Recall that $\cos(A+B) = \cos A\cos B - \sin A\sin B$

$\therefore \cos(A-B) = \cos A\cos B + \sin A\sin B$

$\therefore \cos(A-B) - \cos(A+B) = 2\sin A\sin B$

from which it follows that

$$\langle \sin(nx)\sin(nx)\rangle = \frac{1}{2\pi}\int_0^{2\pi}[\cos[(n-m)x] - \cos[(n+m)x]]dx$$

$$= \frac{1}{2\pi}\left[\frac{\sin[(n-m)x]}{n-m} - \frac{\sin[(n+m)x]}{n+m}\right]_0^{2\pi}$$

At a first glance the above expression simply equals zero.

But now recall from the power series expansion for $\sin x$ that

$\sin x = x + O(x^2)$

from which it follows that:

$$\lim_{n\to m}\frac{\sin[(n-m)x]}{n-m} = \frac{(n-m)x}{n-m} = x$$

$$\lim_{n\to -m}\frac{\sin[(n+m)x]}{n+m} = \frac{(n+m)x}{n+m} = x$$

$$\therefore \langle \sin(nx)\sin(mx)\rangle = \begin{cases} 0, & m=0 \text{ or } n=0 \text{ or } m \neq |n| \\ 1, & m=n \text{ and } n\neq 0 \text{ and } m \neq 0 \\ -1, & m=-n \text{ and } n \neq 0 \text{ and } m \neq 0 \end{cases}$$

5b) $\langle \sin(nx)\cos(mx)\rangle = \dfrac{1}{\pi}\displaystyle\int_0^{2\pi} \sin(nx)\cos(mx)dx$

Recall that $\sin(A+B) = \sin A\cos B + \cos A \sin B$

$\therefore \sin(A-B) = \sin A \cos B - \cos A \sin B$

$\therefore \sin(A+B) + \sin(A-B) = 2\sin A \cos B$

from which it follows that

$$\langle \sin(nx)\cos(nx)\rangle = \frac{1}{2\pi}\int_0^{2\pi} [\sin[(n+m)x] + \sin[(n-m)x]]dx$$

$$= -\frac{1}{2\pi}\left[\frac{\cos[(n+m)x]}{(n+m)} + \frac{\cos[(n-m)x]}{n-m}\right]_0^{2\pi} = 0$$

5c) $\langle \cos(nx)\cos(mx)\rangle = \dfrac{1}{\pi}\displaystyle\int_0^{2\pi} \cos(nx)\cos(mx)dx$

Recall that $\cos(A+B) = \cos A\cos B - \sin A\sin B$

$\therefore \cos(A-B) = \cos A\cos B + \sin A\sin B$

$\therefore \cos(A-B) + \cos(A+B) = 2\cos A\cos B$

from which it follows that

$$\langle \cos(nx)\cos(nx)\rangle = \frac{1}{2\pi}\int_0^{2\pi}[\cos[(n-m)x] + \cos[(n+m)x]]dx$$

$$= \frac{1}{2\pi}\left[\frac{\sin[(n-m)x]}{n-m} + \frac{\sin[(n+m)x]}{n+m}\right]_0^{2\pi}$$

Recalling that

$$\lim_{n\to m}\frac{\sin[(n-m)x]}{n-m} = \frac{(n-m)x}{n-m} = x$$

$$\lim_{n\to -m}\frac{\sin[(n+m)x]}{n+m} = \frac{(n+m)x}{n+m} = x$$

It follows that

$$\langle \cos(nx)\cos(mx)\rangle = \begin{cases} 0, & m \neq |n| \\ 1, & m = |n| \text{ and } n \neq 0 \text{ and } m \neq 0 \\ 2, & m = |n| \text{ and } n = 0 \text{ and } m = 0 \end{cases}$$